

Exact Posterior Probability of Optimality for Factor-Gaussian Beliefs

The Integral That Thompson-Style Methods Approximate,
Evaluated by Quadrature at Scale

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Abstract

The posterior probability that each alternative is best drives stopping rules, budget allocation and Thompson sampling, and is widely treated as intractable. Under common random numbers, and under per-arm Gaussian observations with a factor prior, the posterior covariance is exactly a low-rank factor term plus a diagonal at every step. For such a covariance the whole probability vector is a shared integral over the factors, which quadrature evaluates for all n alternatives at once in time linear in n . In a ranking-and-selection experiment at a common nominal threshold, stopping rules built on marginal means and variances use up to three and a half times as many replicates as the rule built on this vector. In a correlated bandit, sampling from the vector matches Thompson sampling, while a variational policy built on marginals pays more regret.

1 Introduction

Let $\theta \in \mathbb{R}^n$ be the unknown values of n alternatives, such as simulated systems, bandit arms or candidate molecules, and let $\mathcal{D}$ be the data. The posterior probability of optimality is the vector

$$p_i = \Pr\left(\theta_i = \max_j \theta_j \mid \mathcal{D}\right), \quad i = 1, \dots, n. \quad (1)$$

Sequential decision rules use this vector directly. A stopping rule halts when $\max_i p_i$ reaches a confidence target. A budget allocator spends where several p_i are comparable. Thompson sampling plays an action with probability p_i without computing it. A policy with constraints, or one that must be derandomized, needs the vector itself, because one posterior sample does not determine it.

The Bayesian reinforcement learning literature describes Equation (1) as intractable. [Tarbouriech et al. \(2023\)](#) call it the key object of the field and write that computing it “involves computing several complicated integrals with respect to the posterior and is intractable in most cases.” Their method, VAPOR, replaces it with a variational program over occupancy measures.

[Menet et al. \(2026\)](#) fine-tune a large language model toward the same object under a Gaussian-process posterior. The tractable policy they target chooses actions from the marginal means and marginal standard deviations of that posterior alone. Its regret analysis does use the correlation, through a kernel-dependent information-gain bound, but the action probabilities do not.

In ranking and selection, procedures bound the probability of correct selection with Bonferroni-type inequalities or estimate it by posterior sampling. The joint vector is not evaluated.

In several of these settings the posterior covariance has the exact form

$$\Sigma = WW' + \text{diag}(d), \quad W \in \mathbb{R}^{n \times \rho}, \quad \rho \ll n. \quad (2)$$

This form is not a modelling approximation. It follows from how the data are generated. For a covariance of this form, Equation (1) is a shared integral that quadrature evaluates for every i at once, in time linear in n at fixed resolution.

The computation is the pricing pass for correlated contests in Cotton (2026), which also develops its derivatives and its inverse and benchmarks it to $n = 10^6$. This paper contributes three things. Section 3 shows that two standard posteriors keep the form of Equation (2) after every update. Section 4 identifies the approximations in current use and the identity that ties the exact vector to Thompson sampling. Sections 5 and 6 measure what the exact vector changes in a stopping problem and in a bandit.

2 Computing the vector for a factor-form covariance

Write $\theta = m + Wz + \varepsilon$, with $z \sim N(0, I_\rho)$ and independent $\varepsilon_i \sim N(0, d_i)$. Assume every $d_i > 0$. Then the maximum is almost surely unique and the p_i sum to one. Given z , the alternatives are independent, with means $m_i + w'_i z$ and variances d_i . Let $F_{i|z}$ and $f_{i|z}$ be the conditional distribution and density functions. Alternative i is the maximum when every other alternative falls below it, so

$$p_i = \int_{\mathbb{R}^\rho} \phi_\rho(z) \int_{\mathbb{R}} f_{i|z}(x) \prod_{j \neq i} F_{j|z}(x) dx dz. \quad (3)$$

The inner integrands for different i share the product $G_z(x) = \prod_j F_{j|z}(x)$. Dividing out the i th term gives every p_i from the same product:

$$p_i = \mathbb{E}_z \int f_{i|z}(x) \frac{G_z(x)}{F_{i|z}(x)} dx. \quad (4)$$

On a lattice of L points in x , with Q quadrature nodes in z , the whole vector costs $O(nLQ)$. The accuracy depends on L , on the lattice window and on Q , and the Q needed grows with the factor rank and the geometry of W .

The integral is exact, but its evaluation is numerical. With three alternatives, $W = (0, 1, 3)'$ and $d_i = 0.1$, the vector is known analytically from the bivariate quadrant identity, and the default settings of the implementation are in error by total variation 0.0015. Raising the factor quadrature resolution reduces this error.

The numerical details are in Cotton (2026): lattice placement, products taken in logs, and the tails. That implementation computes the probability of being the minimum, and the maximum is obtained by negating m and W . It is used here without other change. The computed vector was compared with argmax frequencies from 2×10^5 Monte Carlo draws at sixteen posteriors: up to four per regime in the stopping experiment, at the twentieth replicate, and two per regime in the bandit, at round 50. The largest total-variation distance in those comparisons is 0.0042, which is at the Monte Carlo noise level for that sample size. These comparisons are statistical checks, not error bounds.

3 Two posteriors with exact factor form

3.1 Common random numbers

Ranking and selection compares n simulated systems. Under common random numbers, replicate r runs every system on the same random scenario:

$$Y_{ir} = \theta_i + v_i' F_r + \varepsilon_{ir}, \quad F_r \sim N(0, I_\rho), \quad \varepsilon_{ir} \sim N(0, \sigma_i^2). \quad (5)$$

The scenario effect F_r is the same for every system. The scenario effects and the noise terms are independent of each other and across replicates. The loadings v_i and noise scales $\sigma_i > 0$ are known properties of the simulation. The unknown is θ , with prior $\theta \sim N(0, s_0^2 I)$ and $s_0 > 0$.

The covariance of one replicate across systems is

$$\Lambda = VV' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_n^2). \quad (6)$$

After $n_r \geq 1$ replicates with mean vector $\bar{Y}$, the posterior of θ is Gaussian with precision

$$A = s_0^{-2} I + n_r \Lambda^{-1}. \quad (7)$$

The Woodbury identity gives

$$\Lambda^{-1} = D^{-1} - D^{-1} V M^{-1} V' D^{-1}, \quad M = I_\rho + V' D^{-1} V. \quad (8)$$

With $U = D^{-1} V$ and $S = n_r M^{-1}$, the precision becomes

$$A = \text{diag}(a) - U S U', \quad a_i = s_0^{-2} + n_r \sigma_i^{-2}. \quad (9)$$

This is a positive diagonal minus a positive semidefinite term of rank at most ρ .

Applying the Woodbury identity again, to Equation (9), gives

$$A^{-1} = \text{diag}(a)^{-1} + \text{diag}(a)^{-1} U C U' \text{diag}(a)^{-1}, \quad C = (S^{-1} - U' \text{diag}(a)^{-1} U)^{-1}. \quad (10)$$

The matrix C is positive definite. To see this, let $K = \begin{pmatrix} \text{diag}(a) & U \\ U' & S^{-1} \end{pmatrix}$. The Schur complement of S^{-1} in K is $\text{diag}(a) - U S U' = A$, which is positive definite, and S^{-1} is positive definite, so K is positive definite. The Schur complement of $\text{diag}(a)$ in K is C^{-1} , so C^{-1} is positive definite too.

The posterior covariance is therefore

$$\Sigma_{\text{post}} = \text{diag}(d) + W W', \quad d_i = a_i^{-1}, \quad W = \text{diag}(a)^{-1} U C^{1/2}, \quad (11)$$

exactly, for every $n_r \geq 1$. Before any replicate the posterior is the diagonal prior. The rank of W is at most ρ , the number of shared scenario streams, and does not grow with the number of systems. The posterior mean is $m = A^{-1} h$ with $h = n_r \Lambda^{-1} \bar{Y}$, and both factors are available from Equations (8) and (10). No step approximates. The posterior has factor form because the scenario noise has factor form.

3.2 Per-arm observations with a factor prior

In a finite Gaussian bandit the correlation sits in the prior on the arm means: $\theta \sim N(0, \Sigma_0)$ with $\Sigma_0 = VV' + \text{diag}(d_0)$, with every $d_{0i} > 0$. Linear models $\theta_i = v_i'\beta + e_i$ with Gaussian β and independent Gaussian e_i produce this form. A Gaussian-process posterior has it only when the approximation replaces the conditional residual by a diagonal. A standard inducing-point posterior has a dense residual (Titsias, 2009), so the diagonal is an additional approximation. Pulling arm a returns $\theta_a + N(0, \sigma_n^2)$, with $\sigma_n > 0$.

With per-arm pull counts c , the posterior precision is

$$A = \Sigma_0^{-1} + \sigma_n^{-2} \text{diag}(c). \quad (12)$$

Equation (8) applied to Σ_0 puts A in the form of Equation (9), with $a_i = d_{0i}^{-1} + c_i\sigma_n^{-2}$, $S = M_0^{-1}$ and $M_0 = I_\rho + V'\text{diag}(d_0)^{-1}V$. The rest of the argument is unchanged. The posterior is exactly factor plus diagonal after every pull, with factor rank no larger than the prior's.

4 Approximations in current use, and Thompson sampling

Two computable rules currently stand in for Equation (1). Both use only the marginal means m_i and marginal standard deviations $s_i = \sqrt{d_i + \|w_i\|^2}$.

F-LITE (Menet et al., 2025) replaces the posterior by $\prod_i N(m_i, s_i^2)$ and approximates the probability of being best by $\Phi((m_i - \kappa^*)/s_i)$, with a threshold κ^* chosen so that these sum to one. The threshold step is a second approximation. With two independent arms, $N(1, 1)$ and $N(0, 1)$, it gives 0.691 for the first arm, against the exact 0.760. F-LITE therefore departs from p even when the posterior is independent.

The *VBOS policy* is the variational Bayesian optimistic sampling rule of O'Donoghue and Lattimore (2021), extended by Tarbouriech et al. (2023). In the near-closed form of Menet et al. (2026) it is

$$\tilde{\pi}_i = v\left(\frac{m_i - \kappa^*}{s_i}\right), \quad v(c) = \exp\left(-\frac{1}{8}(\sqrt{c^2 + 4} - c)^2\right), \quad (13)$$

with κ^* again chosen so the vector sums to one.

The VBOS policy departs from p on purpose, and its regret analysis depends on that departure. It is a policy, not an estimate of p . Its distance from p is reported below because the fine-tuning literature uses it as the trainable substitute for p .

Thompson sampling draws $\tilde{\theta}$ from the posterior and plays the argmax of the draw. Its probability of playing arm i at any history is therefore p_i at that history. Lemma 8 of Tarbouriech et al. (2023) states the general version: the expected occupancy of Thompson sampling equals the posterior probability of optimality.

Sampling an arm from the vector, called *exact matching* below, is therefore Thompson sampling in distribution, up to the quadrature error of Section 2. The two have the same expected regret, and Section 6 compares them. The exact vector adds a computed object, which a stopping rule, an allocator or a derandomized policy can use where a single draw cannot.

5 Stopping under common random numbers

The experiment uses the posterior of Section 3.1 with $n = 100$ systems, $\rho = 2$ scenario streams and prior scale $s_0 = 1$. True means are drawn from the prior. Every system has replicate variance

one, so each replicate carries the same marginal information in every regime. The *factor share* is the fraction of that variance that comes from the shared scenario.

The regimes differ only in how systems load on the scenario streams:

- *Independent*: factor share zero. This is the control.
- *Aligned*: every system loads on the same stream, with factor share 0.6. Differences between systems then vary less than their marginal variances suggest.
- *Opposed*: half the systems load positively and half negatively on one stream, with factor share 0.6. Differences between systems in opposite halves then vary more than their marginal variances suggest.

Three rules compete. Each computes its own estimate of p after every replicate, stops the first time its largest entry reaches $1 - \delta$ with $\delta = 0.05$, and selects that entry. The exact rule uses the computation of Section 2. The other two use F-LITE and the VBOS policy.

For the true vector the rule has a Bayesian guarantee. Let τ be the first time $\max_i p_i \geq 1 - \delta$, let B be the budget, and on $\{\tau \leq B\}$ let I be the selected alternative and I^* the best. Then

$$\Pr(I \neq I^* \mid \tau \leq B) = \mathbb{E}[1 - p_I(\mathcal{D}_\tau) \mid \tau \leq B] \leq \delta. \quad (14)$$

Optional stopping does not affect this statement, which averages over the prior. It is not a frequentist guarantee for a fixed θ , and it applies to the true p , not to a numerical approximation of it.

The three rules share a nominal threshold, not a true confidence level. With two independent arms of equal posterior standard deviation s , an entry of 0.95 requires a mean gap of 2.33 s for the exact vector, 3.29 s for F-LITE and 4.84 s for the VBOS policy. The true posterior probability of a correct selection at those gaps is 0.950, 0.990 and 0.9997. Part of any extra sampling by the marginal rules is this higher effective target.

regime	rule	mean replicates	realized PCS	censored of 200
independent	exact	141.2	0.970	35
	F-LITE	207.0	0.985	65
	VBOS	271.5	1.000	102
aligned	exact	69.7	0.958	9
	F-LITE	175.6	1.000	60
	VBOS	240.1	1.000	91
opposed	exact	106.3	0.944	21
	F-LITE	171.7	0.979	57
	VBOS	239.0	0.991	88

Table 1: Stopping at nominal 95% confidence. Each regime has 200 replications and a budget of 400 replicates. Mean replicates is the mean of the stopping time capped at 400, so a censored run counts 400. Realized PCS is the fraction of stopped runs that selected the true best system. The stopped runs differ by rule.

In Table 1, every rule’s realized PCS among its stopped runs is at or above nominal, within binomial noise. The exact rule’s 0.944 in the opposed regime has 179 stopped runs and a standard error of 0.016.

At this common nominal threshold and budget, the two marginal-only rules used 1.5 to 3.4 times the capped mean replicates of the exact rule, and hit the budget far more often. The gap is largest in the aligned regime, where the leading systems’ differences are tightly coupled. There the exact rule stops after 69.7 replicates on average, against 175.6 and 240.1, and stops in 191 of 200 runs, against 140 and 109. An efficiency comparison at a matched true error would need thresholds calibrated to a common PCS, which this experiment does not do.

The marginal-only estimates are far from the exact vector along the way. Before stopping in the aligned regime, F-LITE is at median total-variation distance 0.123 from the exact vector, with ninetieth percentile 0.174. The VBOS policy is at 0.141, with ninetieth percentile 0.206.

The distances do not vanish in the independent regime, where they are 0.063 and 0.096. F-LITE’s threshold step is biased even when the posterior is independent, and the VBOS policy departs from p by design. At the stopping experiment’s Monte Carlo checks, the largest total-variation distance of the computed vector was 0.0015.

6 A correlated Gaussian bandit

The bandit uses the posterior of Section 3.2 with $n = 50$ arms, prior variance one per arm, observation noise $\sigma_n = 1$, horizon 200 and 100 replications. The prior’s factor term has two columns. In the *independent* regime it is zero. In the *clusters* regime each half of the arms loads on its own factor, which carries 0.75 of the prior variance.

Four policies act on the same posterior: Thompson sampling, exact matching, sampling from the VBOS policy, and sampling from F-LITE. Table 2 reports cumulative Bayes regret and, at the horizon, whether the arm with the highest posterior mean is the true best and the shortfall of that arm’s value below the best. Its last column is the total-variation distance between each policy’s action distribution and the exact vector at mid-horizon, along that policy’s own trajectory.

regime	policy	regret	identification	shortfall	TV to exact p
independent	Thompson	172.1	0.72	0.073	0
	exact matching	173.9	0.75	0.055	0
	VBOS	190.3	0.76	0.061	0.159
	F-LITE	179.2	0.79	0.052	0.057
clusters	Thompson	122.7	0.61	0.097	0
	exact matching	124.8	0.51	0.113	0
	VBOS	135.9	0.59	0.105	0.129
	F-LITE	128.7	0.51	0.121	0.061

Table 2: Mean over 100 replications of cumulative Bayes regret, the rate at which the highest posterior mean identifies the true best arm at the horizon, and the shortfall of that arm. The last column is the total-variation distance from the computed vector at mid-horizon. It is zero for exact matching by construction. For Thompson sampling it is zero relative to the true p by Lemma 8, and is entered as zero rather than measured.

Thompson sampling and exact matching have the same expected regret by Lemma 8, and their sample means differ by one to two percent of cumulative regret in both regimes. The VBOS policy’s mean regret is ten to eleven percent above Thompson sampling’s, and its action

distribution is at total-variation distance 0.13 to 0.16 from the computed vector. F-LITE’s is four to five percent above. Per-replication regrets were not saved, so these differences carry no paired intervals.

The identification rates do not separate the policies. Their differences are within two binomial standard errors at 100 replications. The regret and distance columns carry the comparison.

7 Scope

The derivations of Section 3 are exact because the observations there are generated with factor structure. When scenario effects enter a simulation nonlinearly, a low-rank model of Λ is an approximation, and the posterior inherits its error. When a Gaussian-process kernel is dense with a short length scale, a factor fit has to precede the computation, and its residual is the cost.

The probability of optimality is not the knowledge gradient. Under simple-regret loss, the one-step value of a measurement is the knowledge-gradient quantity (Frazier et al., 2008), an expectation over the next posterior means. Tuisov (2026) shows that this is the normative myopic value of computation, and notes that the exact dynamic version “is itself generally intractable.” Neither quantity is computed here.

The regret analysis of O’Donoghue and Lattimore (2021) depends on the VBOS tilt. Its measured distance from p quantifies the approximation, but does not by itself make a case against the tilt.

Thompson sampling needs one posterior sample and no vector. The exact vector matters where a sample is not enough: stopping, allocation, constraints and derandomization. Both experiments above are of that kind.

References

- P. Cotton. Scalable inversion of contests with correlated performances, including softmax and multinomial probit. arXiv:2609.01133, 2026. Also SSRN 7307363.
- P. I. Frazier, W. B. Powell, and S. Dayanik. A knowledge-gradient policy for sequential information collection. *SIAM Journal on Control and Optimization*, 47(5):2410–2439, 2008.
- N. Menet, J. Huebotter, P. Kassraie, and A. Krause. Efficiently estimating Gaussian probability of maximality. In *AISTATS*, 2025. arXiv:2501.13535.
- N. Menet, A. Terzić, M. Hersche, A. Krause, and A. Rahimi. Thompson sampling via fine-tuning of LLMs. In *ICLR*, 2026. arXiv:2510.13328.
- B. O’Donoghue and T. Lattimore. Variational Bayesian optimistic sampling. In *Advances in Neural Information Processing Systems 34*, 2021. arXiv:2110.15688.
- J. Tarbouriech, T. Lattimore, and B. O’Donoghue. Probabilistic inference in reinforcement learning done right. In *Advances in Neural Information Processing Systems 36*, 2023. arXiv:2311.13294.
- M. Titsias. Variational learning of inducing variables in sparse Gaussian processes. In *AISTATS*, 2009.
- A. Tuisov. Search as computation allocation. arXiv:2607.27871, 2026.